PADHAKU
Science · Class 10 Science



CHAPTER 9

# Light – Reflection and Refraction

NCERT Exercise Solutions


| | |
|---|---|
| Activity · 9.1 Reflection of Light | 1 Q |
| Activities · 9.2 Spherical Mirrors | 5 Q |
| Questions · 9.2 Spherical Mirrors | 4 Q |
| Questions · 9.2.4 Mirror Formula and Magnification | 2 Q |
| Activities · 9.3 Refraction of Light | 4 Q |
| Questions · 9.3.2 The Refractive Index | 5 Q |
| Activities · 9.3.3–9.3.4 Lenses | 3 Q |
| Questions · 9.3.8 Power of a Lens | 3 Q |
| Exercises | 17 Q |


✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

## Activity · 9.1 Reflection of Light

**Activity 9.1** **Take a large shining spoon and view your face in its curved surface. Is the image smaller or larger? Move the spoon away – how does the image change? Reverse the spoon and repeat. Compare the images on the two surfaces.**

✔ ANSWER

**Inner (curved inwards) surface – acts as a concave mirror:**

- Very close to the face: the image is erect and larger.
- As the spoon moves away: the image becomes inverted and smaller.

**Outer (bulging) surface – acts as a convex mirror:**

- The image is always erect and smaller; it gets still smaller as the spoon moves away.

**Conclusion:** A curved mirror can form images of different size and nature; the inner surface behaves like a concave mirror and the outer like a convex mirror.

## Activities · 9.2 Spherical Mirrors

**Activity 9.2** **Direct a concave mirror towards the Sun and focus the reflected light on a sheet of paper to get a bright, sharp**

**spot. Hold them in place for a few minutes. What do you observe? Why?**

✔ ANSWER

**Observation:** The paper starts to burn, gives smoke and may even catch fire.

**Reason:** The concave mirror converges the parallel rays of sunlight to one point – its principal focus. The bright spot is the real image of the Sun. The concentrated sunlight produces heat, which burns the paper.

The distance of the spot from the mirror is the approximate focal length of the mirror.

Activity 9.3

**Place a candle in front of a concave mirror: far beyond C, (a) just beyond C, (b) at C, (c) between F and C, (d) at F, (e) between P and F. Note the nature, position and size of the image. In which case is no image obtained on the screen?**

✔ ANSWER

| Position of object | Position of image | Size | Nature |
|---|---|---|---|
| Far beyond C (at infinity) | At F | Highly diminished, point-sized | Real, inverted |
| Just beyond C | Between F and C | Diminished | Real, inverted |
| At C | At C | Same size | Real, inverted |

| | | | |
|---|---|---|---|
| Between F and C | Beyond C | Enlarged | Real, inverted |
| At F | At infinity | Highly enlarged | Real, inverted |
| Between P and F | Behind the mirror | Enlarged | Virtual, erect |

When the candle is **between P and F**, no image forms on the screen; a virtual, erect, enlarged image is seen in the mirror itself.

**Activity 9.4** **Draw ray diagrams for each position of the object in Table 9.1 using any two standard rays, compare with Fig. 9.7, and tabulate the nature, position and size of the image.**

💡 SAMPLE ANSWER

Use any two of these rays: (i) a ray parallel to the principal axis passes through F after reflection; (ii) a ray through F becomes parallel to the axis; (iii) a ray through C comes back along the same path; (iv) a ray hitting P is reflected making an equal angle with the axis.

The point where the reflected rays meet (or appear to meet) is the image. The results match Table 9.1 (see the table in Activity 9.3 above).

(Draw the diagrams as in Fig. 9.7 of your book.)

**Activity 9.5** **Observe the image of an upright pencil in a convex mirror. Is it erect or inverted? Diminished or enlarged? Move the pencil away – does the image become smaller or larger? Does the image move closer to or farther from the focus?**

✔ ANSWER

**Observation:** The image is always **erect** and **diminished**.

As the pencil moves away, the image becomes **smaller** and moves **closer to the focus** F (behind the mirror).

**Conclusion:** A convex mirror always forms a virtual, erect, diminished image between P and F.

**Activity 9.6** **Observe the image of a distant tall object (a tree) in a plane mirror, a concave mirror and a convex mirror. In which do you see the full-length image? Explain.**

✔ ANSWER

- **Plane mirror:** a small plane mirror shows only part of the tree; a full image needs a very large mirror (image size = object size).
- **Concave mirror:** the image of a distant object is inverted and small, near the focus; it is not easy to see the full object clearly.

- **Convex mirror:** shows the **full-length** erect image even in a small mirror.

**Reason:** A convex mirror is curved outwards, so it has a wider field of view and always forms a diminished, erect image.

## ❓ Questions · 9.2 Spherical Mirrors

**Q 1** **Define the principal focus of a concave mirror.**

✔ ANSWER

The principal focus of a concave mirror is the point on its principal axis where rays of light parallel to the principal axis **meet after reflection** from the mirror.

**Q 2** **The radius of curvature of a spherical mirror is 20 cm. What is its focal length?**

✔ ANSWER

Given: R = 20 cm

f = R/2 = 20/2

**f = 10 cm**

**Q 3** **Name a mirror that can give an erect and enlarged image of an object.**

✔ ANSWER

A **concave mirror**, when the object is placed between its pole (P) and principal focus (F).

**Q 4** **Why do we prefer a convex mirror as a rear-view mirror in vehicles?**

✔ ANSWER

- A convex mirror always gives an **erect**, though diminished, image.
- It is curved outwards, so it has a **wider field of view**: the driver can see a much larger area of traffic behind than with a plane mirror.

## Questions · 9.2.4 Mirror Formula and Magnification

**Q 1** **Find the focal length of a convex mirror whose radius of curvature is 32 cm.**

✔ ANSWER

Given: R = +32 cm (convex mirror, centre of curvature behind the mirror)

f = R/2 = +32/2

**f = +16 cm** (16 cm behind the mirror)

**Q 2** **A concave mirror produces three times magnified (enlarged) real image of an object placed at 10 cm in front of it. Where is the image located?**

✔ ANSWER

Given: u = −10 cm; real image, so m = −3

m = −v/u

−3 = −v/(−10)

v = −30 cm

**The image is 30 cm in front of the mirror** (on the same side as the object). It is real and inverted.

## ⭐ Activities · 9.3 Refraction of Light

**Activity 9.7** **Place a coin at the bottom of a bucket of water. With your eye to a side above the water, try to pick it up in one go. Did you succeed? Why not?**

✔ ANSWER

**Observation:** Usually we do not succeed in one go; the hand reaches the wrong place.

**Reason:** Light from the coin bends (refracts) when it passes from water into air. So the coin appears raised and shifted from its real position, and we aim at its image, not the coin.

**Activity 9.8** **Put a coin in a shallow bowl and move back till the coin just disappears. A friend pours water gently into the bowl. Does the coin become visible again? How?**

✔ ANSWER

**Observation:** Yes, the coin becomes visible again.

**Reason:** Light from the coin bends away from the normal as it goes from water into air, and reaches our eye. The coin appears slightly raised above its actual position due to **refraction of light**.

**Activity 9.9** **Place a glass slab over a thick ink line so that its edge makes an angle with the line, then normal to the line. Does the line appear bent at the edges? Seen from the top, does the line under the slab appear raised? Why?**

✔ ANSWER

- Slab edge at an angle to the line: the part of the line under the slab appears **bent (shifted)** at the edges.
- Slab normal to the line: the line does **not** appear bent (light falling along the normal is not bent).
- From the top: the line under the slab appears **raised**.

**Reason:** Light from the line bends when it goes from glass into air (refraction).

**Activity 9.10** **Trace the path of a ray through a rectangular glass slab ABCD using pins E, F, G, H (Fig. 9.10). Compare the angles of incidence and refraction at AB and CD. What happens when a ray falls normally on the slab?**

✔ ANSWER

- At O (air → glass, rarer → denser) the ray bends **towards the normal**: angle of refraction < angle of incidence.
- At O′ (glass → air, denser → rarer) the ray bends **away from the normal**: angle of emergence > angle of refraction.
- The emergent ray O′H is **parallel** to the incident ray EO, but shifted sideways slightly (lateral displacement), because the bending at the two parallel faces is equal and opposite.
- A ray falling **normally** (along the normal) on the slab goes straight through **without bending**.

## ❓ Questions · 9.3.2 The Refractive Index

**Q 1** **A ray of light travelling in air enters obliquely into water. Does the light ray bend towards the normal or away from the normal? Why?**

✔ ANSWER

It bends **towards the normal**.

Water is optically denser than air (n = 1.33), so light slows down in water and bends towards the normal.

**Q 2** **Light enters from air to glass having refractive index 1.50. What is the speed of light in the glass? The speed of light in vacuum is $3 \times 10^8$ m $s^{-1}$.**

✔ ANSWER

Given: n = 1.50, c = $3 \times 10^8$ m $s^{-1}$

n = c/v, so v = c/n

v = $(3 \times 10^8)/1.50$

**v = $2 \times 10^8$ m $s^{-1}$**

**Q 3** **Find out, from Table 9.3, the medium having highest optical density. Also find the medium with lowest optical density.**

✔ ANSWER

Highest optical density: **diamond** (n = 2.42).

Lowest optical density: **air** (n = 1.0003).

**Q 4** **You are given kerosene, turpentine and water. In which of these does the light travel fastest? Use the information given in Table 9.3.**

✔ ANSWER

Refractive indices: water 1.33, kerosene 1.44, turpentine oil 1.47.

Light is fastest in the medium with the lowest refractive index, so it travels fastest in **water**.

**Q 5** **The refractive index of diamond is 2.42. What is the meaning of this statement?**

✔ ANSWER

It means the ratio of the speed of light in air (vacuum) to the speed of light in diamond is 2.42.

So light travels **2.42 times slower** in diamond than in vacuum ($v = c/2.42 \approx 1.24 \times 10^8$ m $s^{-1}$).

## ⭐ Activities · 9.3.3–9.3.4 Lenses

**Activity 9.11** **Direct a convex lens towards the Sun and focus the light on a sheet of paper. Hold them in place for a while. What happened? Why?**

✔ ANSWER

**Observation:** The paper begins to burn, gives smoke and may catch fire.

**Reason:** Sunlight consists of parallel rays. The convex lens converges them at its principal focus; the bright spot is the real image of the Sun. The concentrated sunlight produces heat that burns the paper (just like the concave mirror in Activity 9.2).

The distance between the lens and the spot is the approximate focal length of the lens.

**Activity 9.12** **Place a candle in front of a convex lens: far beyond $2F_1$, just beyond $2F_1$, at $2F_1$, between $F_1$ and $2F_1$, at $F_1$, between $F_1$ and O. Note and tabulate the nature, position and size of the image.**

✔ ANSWER

| Position of object | Position of image | Size | Nature |
|---|---|---|---|
| At infinity (far beyond $2F_1$) | At $F_2$ | Highly diminished, point-sized | Real, inverted |
| Beyond $2F_1$ | Between $F_2$ and $2F_2$ | Diminished | Real, inverted |
| At $2F_1$ | At $2F_2$ | Same size | Real, inverted |
| Between $F_1$ and $2F_1$ | Beyond $2F_2$ | Enlarged | Real, inverted |
| At $F_1$ | At infinity | Highly enlarged | Real, inverted |

| | | | |
|---|---|---|---|
| Between $F_1$ and O | Same side as the object | Enlarged | Virtual, erect |

When the candle is between $F_1$ and O, no image forms on the screen; the virtual image is seen by looking through the lens.

**Activity 9.13** **Observe the image of a candle through a concave lens. Note its nature, size and position. Move the candle away. What happens to the size of the image when the candle is very far from the lens?**

✔ ANSWER

**Observation:** The image cannot be obtained on a screen. Seen through the lens it is **virtual, erect and diminished**, on the same side as the candle, between $F_1$ and O.

As the candle moves away, the image becomes **smaller**. When the candle is very far (at infinity), the image is highly diminished (point-sized) at the focus $F_1$.

**Conclusion:** A concave lens always gives a virtual, erect and diminished image, whatever the position of the object.

## ❓ Questions · 9.3.8 Power of a Lens

**Q 1** **Define 1 dioptre of power of a lens.**

✔ ANSWER

**1 dioptre** is the power of a lens whose focal length is **1 metre**. 1 D = 1 $m^{-1}$.

**Q 2** **A convex lens forms a real and inverted image of a needle at a distance of 50 cm from it. Where is the needle placed in front of the convex lens if the image is equal to the size of the object? Also, find the power of the lens.**

✔ ANSWER

A real image of the same size forms when the object is at $2F_1$ and the image at $2F_2$.

So v = +50 cm and u = −50 cm. **The needle is 50 cm in front of the lens.**

Lens formula: 1/f = 1/v − 1/u = 1/50 − 1/(−50) = 2/50 = 1/25

f = +25 cm = +0.25 m

P = 1/f = 1/0.25

**P = +4 D**

**Q 3** **Find the power of a concave lens of focal length 2 m.**

✔ ANSWER

Given: f = −2 m (concave lens)

P = 1/f = 1/(−2)

**P = −0.5 D**

## 📝 Exercises

**1** **Which one of the following materials cannot be used to make a lens?**
**(a) Water (b) Glass (c) Plastic (d) Clay**

✔ ANSWER

**(d) Clay** – it is opaque; a lens must be made of a transparent material.

**2** **The image formed by a concave mirror is observed to be virtual, erect and larger than the object. Where should be the position of the object?**
**(a) Between the principal focus and the centre of curvature**
**(b) At the centre of curvature**
**(c) Beyond the centre of curvature**
**(d) Between the pole of the mirror and its principal focus**

✔ ANSWER

**(d) Between the pole of the mirror and its principal focus** – only in this position does a concave mirror form a virtual, erect, enlarged image.

**3** **Where should an object be placed in front of a convex lens to get a real image of the size of the object?**

**(a) At the principal focus of the lens**

**(b) At twice the focal length**

**(c) At infinity**

**(d) Between the optical centre of the lens and its principal focus**

✔ ANSWER

**(b) At twice the focal length** (at $2F_1$) – the image is then real, inverted, same size, at $2F_2$.

**4** **A spherical mirror and a thin spherical lens have each a focal length of –15 cm. The mirror and the lens are likely to be**

**(a) both concave**

**(b) both convex**

**(c) the mirror is concave and the lens is convex**

**(d) the mirror is convex, but the lens is concave**

✔ ANSWER

**(a) both concave** – by the sign convention, a negative focal length means a concave mirror and a concave lens.

**5** **No matter how far you stand from a mirror, your image appears erect. The mirror is likely**

to be

(a) only plane (b) only concave (c) only convex (d) either plane or convex

✔ ANSWER

**(d) either plane or convex** – both always form erect (virtual) images for any object distance.

6 **Which of the following lenses would you prefer to use while reading small letters found in a dictionary?**

**(a) A convex lens of focal length 50 cm**

**(b) A concave lens of focal length 50 cm**

**(c) A convex lens of focal length 5 cm**

**(d) A concave lens of focal length 5 cm**

✔ ANSWER

**(c) A convex lens of focal length 5 cm** – a convex lens gives an enlarged, erect image when the object is within its focus, and a short focal length gives greater magnification.

7 **We wish to obtain an erect image of an object, using a concave mirror of focal length 15 cm. What should be the range of distance of the object from the mirror? What is the nature of the image? Is the image larger or smaller than the object? Draw a ray diagram to show the image formation in this case.**

✔ ANSWER

- Range: the object must be between the pole and the focus, i.e. **between 0 and 15 cm** from the mirror.
- Nature: **virtual and erect**, formed behind the mirror.
- Size: **larger** than the object (enlarged).

**Ray diagram:** Draw the object AB between P and F. From B draw (1) a ray parallel to the principal axis that reflects through F, and (2) a ray towards P that reflects at an equal angle with the axis (or a ray along the line from C through B, which returns on itself). The reflected rays diverge; extend them backwards (dotted) behind the mirror. They meet at B′, giving an enlarged, erect image A′B′ behind the mirror. (See the last diagram of Fig. 9.7 in your book.)

**8** **Name the type of mirror used in the following situations. Support your answer with reason.**

**(a) Headlights of a car**

**(b) Side/rear-view mirror of a vehicle**

**(c) Solar furnace**

✔ ANSWER

(a) **Concave mirror** – the bulb is placed at its focus, so the reflected light goes out as a powerful parallel beam.

(b) **Convex mirror** – it always gives an erect, diminished image and has a wider field of view, so the driver sees a large area behind.

(c) **Concave mirror** – a large concave mirror converges sunlight at its focus, concentrating heat to produce high temperature.

**9** **One-half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object? Verify your answer experimentally. Explain your observations.**

✔ ANSWER

**Yes**, the lens still forms a **complete image**, but it is **less bright** (dimmer).

**Experiment:** Cover the lower half of a convex lens with black paper. Place a burning candle beyond $2F_1$ and get its image on a screen on the other side. The full image of the flame is obtained, but fainter than with the uncovered lens.

**Explanation:** Rays from every point of the object still pass through the uncovered half and meet at the image point. Since only half of the light passes, the image is less bright.

**10** **An object 5 cm in length is held 25 cm away from a converging lens of focal length 10 cm. Draw the ray diagram and find the position, size and the nature of the image formed.**

✔ ANSWER

Given: h = +5 cm, u = −25 cm, f = +10 cm (convex lens)

Lens formula: 1/v − 1/u = 1/f

1/v = 1/f + 1/u = 1/10 + 1/(−25) = 1/10 − 1/25

1/v = (5 − 2)/50 = 3/50

v = 50/3 = **+16.7 cm** (on the other side of the lens)

m = v/u = (+16.7)/(−25) = **−0.67**

h′ = m × h = −0.67 × 5 = **−3.3 cm**

**Nature:** real, inverted and diminished, 16.7 cm behind the lens (between $F_2$ and $2F_2$).

**Ray diagram:** Object beyond $2F_1$. A ray from the top parallel to the axis refracts through $F_2$; a ray through O goes straight. They meet between $F_2$ and $2F_2$ below the axis, giving a small inverted image. (See Fig. 9.16, object beyond $2F_1$.)

**11** **A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed from the lens? Draw the ray diagram.**

✔ ANSWER

Given: f = −15 cm, v = −10 cm (concave lens: virtual image on the same side as the object)

Lens formula: 1/v − 1/u = 1/f, so 1/u = 1/v − 1/f

1/u = 1/(−10) − 1/(−15) = −1/10 + 1/15

1/u = (−3 + 2)/30 = −1/30

**u = −30 cm**: the object is 30 cm from the lens, on the same side as the image.

**Ray diagram:** Object 30 cm to the left. A ray from the top parallel to the axis refracts and appears to come from $F_1$ (extend it back, dotted); a ray through O goes straight. They appear to meet at 10 cm from the lens, giving a virtual, erect, diminished image between $F_1$ and O. (See Fig. 9.17.)

**12** **An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.**

✔ ANSWER

Given: u = −10 cm, f = +15 cm (convex mirror)

Mirror formula: 1/v + 1/u = 1/f, so 1/v = 1/f − 1/u

1/v = 1/15 − 1/(−10) = 1/15 + 1/10

1/v = (2 + 3)/30 = 5/30 = 1/6

**v = +6 cm**: the image is 6 cm **behind** the mirror.

m = −v/u = −(6)/(−10) = +0.6

**Nature:** virtual, erect and diminished.

**13** **The magnification produced by a plane mirror is +1. What does this mean?**

✔ ANSWER

- m = 1: the image is the **same size** as the object.

- Positive sign: the image is **virtual and erect**.

**14** **An object 5.0 cm in length is placed at a distance of 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position of the image, its nature and size.**

✔ ANSWER

Given: h = +5.0 cm, u = −20 cm, R = +30 cm, so f = R/2 = +15 cm

1/v = 1/f − 1/u = 1/15 − 1/(−20) = 1/15 + 1/20

1/v = (4 + 3)/60 = 7/60

v = 60/7 = **+8.6 cm** (behind the mirror)

m = −v/u = −(8.6)/(−20) = +0.43

h′ = m × h = 0.43 × 5.0 = **+2.1 cm**

**Nature:** virtual, erect and diminished (2.1 cm tall), 8.6 cm behind the mirror.

**15** **An object of size 7.0 cm is placed at 27 cm in front of a concave mirror of focal length 18 cm. At what distance from the mirror should a screen be placed, so that a sharp focussed image can be obtained? Find the size and the nature of the image.**

✔ ANSWER

Given: h = +7.0 cm, u = −27 cm, f = −18 cm (concave mirror)

1/v = 1/f − 1/u = 1/(−18) − 1/(−27) = −1/18 + 1/27

1/v = (−3 + 2)/54 = −1/54

**v = −54 cm**: place the screen **54 cm in front of the mirror**.

m = −v/u = −(−54)/(−27) = −2

h′ = m × h = −2 × 7.0 = **−14 cm**

**Nature:** real, inverted and enlarged (14 cm tall).

**16** **Find the focal length of a lens of power − 2.0 D. What type of lens is this?**

✔ ANSWER

f = 1/P = 1/(−2.0) m

**f = −0.5 m = −50 cm**

Negative focal length, so it is a **concave (diverging) lens**.

**17** **A doctor has prescribed a corrective lens of power +1.5 D. Find the focal length of the lens. Is the prescribed lens diverging or converging?**

✔ ANSWER

f = 1/P = 1/1.5 m

**f = +0.67 m ≈ +66.7 cm**

Positive power, so it is a convex lens – a **converging** lens.

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**